

jouleos

The search for JouleOS

David Charlot, Open Interface Engineering

Research study · OpenIE research.openie.dev

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Measurement. This study meters nothing. Figures quoted from a project are that project's own claims. An estimator is not a measured joule.

Existing compute fabric treats the floor as fixed. Kernels, builds, guests, and instruction contracts are collapsed into one rung. Agents are wrapped on top of that settlement. Energy to run is not the condition of a commit. Secure compute requires one law, one meter, and one refusal. That requirement is the search for a solution like JouleOS.

OpenIE research study five. Written 3 October 2026 (ET).

Computers are hardware. Software is applied engineering written for that hardware under constraints. Energy to run is the only true metric of computer intelligence. All other factors collapse to zero.

1. What this study is

This study is the account of existing compute fabric and the software written for it. The problems of that fabric and that software are the point. A machine that schedules, meters, and refuses an agent under law does not exist on the stack as it is composed today. The need those problems state is a solution of the JouleOS kind. The study is not a document of that project.

It is a review of how secure compute is composed today. It is a design guide for what a machine must speak if an agent is to be scheduled, metered, and refused under law. It is not a ranking of kernels. It is not a slogan about green computing.

The OpenIE thesis is stated once and held for the rest of the file. Energy to run is the only true metric of computer intelligence. Latency, token counts, accuracy benches, and brand names are not the unit that decides whether a commit is allowed. They collapse when the meter and the refusal are real. Mixture of Limits is the schedule, the meter, and the refusal on that machine. Metabolic Intelligence is the envelope of the best answer that fits the energy the hardware can spend. Neither name is abbreviated in this draft.

The review depends on an internal working note that drew the unsettled stack before any destination was named: the OpenIE stack working note, with the sift in the OpenIE convergence working note. Those files are working notes. They are not published papers. Where a fact lives only there, this draft cites the working note and does not pretend a journal published it.

The standards that name the schedule, the law, the commit, and the isolation ladder are the Transaction Science page sources and the JouleContract draft: the *-transaction-science-web trees, and JouleContract draft 0.3. JouleOS is named where a solution of that kind is the answer the problems require.

This study does not invent measured joules. When the JouleOS README states Lawful near one microjoule, LlmInLoop near five millijoules, or a roughly one-hundred-thousand-fold ratio against an LLM-everything corpus, those figures are quoted as that project's own claims and labeled as such. Estimator readings are not measured joules. Nominal ranges drawn for application pillars on a standards site are not metered joules and are not twenty-two operating systems.

2. The view that assumes the floor is fixed

Many projects make a view of the agent-safety problem while assuming a layer underneath is fixed. The kernel stays. The process model stays. The install path stays. The agent is wrapped. The wrap is the product. Two concrete cases already in the source material show the class.

2.1 NVIDIA OpenShell on top of an existing agent

On 28 September 2026 NVIDIA announced the Open Agent Safety Platform. OpenShell is the open-source secure runtime in that platform. The public technical blog and investor release describe a sandbox, a supervisor outside the agent workload, policy controls, and an optional out-of-band watchdog named Sentry that can run on BlueField data processing units. The agent remains a child. The host operating system and its privileged law remain the floor. OpenShell sets boundaries for agents running on central processors. Sentry adds monitoring and quarantine from a separate silicon domain when BlueField hardware is present. OpenShell can run without that hardware. The class mechanism is clear: safety is a runtime layered onto an existing agent stack, not a rewrite of the machine's law.

Sources for this placement: NVIDIA Technical Blog, "Add Runtime Controls to AI Agents with NVIDIA OpenShell," 28 September 2026 (<https://developer.nvidia.com/blog/add-runtime-controls-to-ai-agents-with-nvidia-openshell/>); NVIDIA investor press release dated 28 September 2026 (<https://investor.nvidia.com/news/press-release-details/2026/NVIDIA-Launches-Open-Agent-Safety-Platform-to-Secure-Agents-From-Testing-to-Deployment/default.aspx>); NVIDIA OpenShell documentation on how the gateway, supervisor, and sandbox split control and enforcement (<https://docs.nvidia.com/openshell/dev/about/how-it-works>).

2.2 OpenHuman: a Rust harness whose core stays on the host

OpenHuman is an open-source agent harness with a Rust core, licensed under GPL-3.0, and described by its README as early beta (<https://github.com/tinyhumansai/openhuman>). The same core runs a desktop app, a browser UI, a terminal client, and an embeddable library. The density figures in that README are memory measurements from the project's own fleet sweeps and cold-start timings. They are not joules. This study records them only as the project's memory claims and does not convert them into energy.

The sandbox design in the project's own pull requests and security docs is the class mechanism that matters here. Sandbox backends are named None, Local, and Docker. Local uses Landlock on Linux, Seatbelt on macOS, or AppContainer on Windows, with a documented no-op fallback where the host provides no jail. Docker runs ephemeral containers with hardened defaults when that path is selected. ReadOnly resolves to None for execution enforcement: it does not add a jail. Elevated tools such as git, install, docker management, and process operations are defined to leave the jail and run on the host path. The core always runs on the host. The agent is confined where the harness chooses to confine tool execution. The host kernel and process model remain the floor.

Sources: OpenHuman README (<https://raw.githubusercontent.com/tinyhumansai/openhuman/main/README.md>); OpenHuman pull request 3261 describing sandbox backends and the elevated path (<https://github.com/tinyhumansai/openhuman/pull/3261>); OpenHuman security architecture page (<https://tinyhumans.gitbook.io/openhuman/developing/architecture/security>).

2.3 The class, not the brand

OpenShell and OpenHuman are not the only members of the class. They are two named instances. The class mechanism is the same. A view of agent safety is built while the privileged software under the agent is treated as settled. Mixture of Limits cannot be only a supervisor beside that settlement. Metabolic Intelligence cannot be only a cheaper model behind the same install path. The floor has to be part of the review.

3. The teaching line: silicon to human interface

Before any product name, teach the line. The working note calls it the gospel stack. It is a teaching order, not a census of products. Bottom to top:

1. Silicon
2. Chip
3. Board
4. Firmware
5. Instruction set
6. Hardware abstraction
7. Operating system
8. Compiler and runtime

- 9. Programs
- 10. Human interface

Silicon is the material. The chip is that circuit as a part. The board is the chip with power, clocks, memory, and connectors. Those three are the computer. They exist whether or not any program is present.

Firmware is the first software written for that board. It brings the board up and talks to the devices on it. It is applied engineering. It is not a layer of the silicon.

The instruction set is the contract the chip offers. It names the operations the chip will perform and the rules for registers, memory, and traps. Assembly sits beside that contract. It is text that names the chip's operations directly. On a host computer a compiler usually emits it. On a microcontroller it is often where the software product stops. The teaching line does not invent an eleventh numbered rung for assembly. The figure in the working note marks assembly beside the instruction set.

Hardware abstraction is software that hides board and chip differences from the software above it. Vendor documents do not all place it on a free-floating rung under the operating system. Microsoft's driver documentation places a hardware abstraction layer inside Windows. CMSIS and a vendor HAL such as the STM32CubeF4 HAL are libraries, not operating systems. The teaching line keeps David's order. The later contracts refuse to promote hardware abstraction into a seventh kind of thing.

An operating system, on this line, is software that manages the computer for other programs. That rung is not one job. Section 4 of the working study takes it apart. This draft will do the same by job, not by fame.

A compiler turns a person's text into instructions for a contract. A runtime stays with the program while it runs. Both are applied engineering for a hardware contract. They are not the computer.

Programs are the work. An agent is a program. It occupies this rung. It is not a new kind of matter and not an operating system.

The human interface is where a person meets that work: a screen, a keyboard, a pointer, a voice path, or a serial console. The teaching line ends at a person. Agents, sensors, and other machines are real clients of that computer. They are not rungs above the person. They are other stacks, or guests of a stack.

Source for the ten-rung order and the placements above: the OpenIE stack working note sections 2, 3, and 7, with figures 01–gospel–stack and 02–where–machines–stop.

4. Where the teaching line already splits

The line looks like one ladder. It already splits three ways in the working study.

4.1 Host computer and microcontroller

A microcontroller often stops before a host operating system. Its honest stops are firmware and assembly alone, or a small real-time kernel that is not a host product. FreeRTOS and Zephyr are kernels for that region. seL4's own questions page says using seL4 without a full memory management unit makes little sense, because its resource management is based on virtual memory. Different machines. Different stops. One teaching line.

4.2 Assembly and the compiler

Assembly and a compiler are not two names for one rung. On the microcontroller line, the compiler often ran on a host and its output was loaded onto the board. On the host line, the compiler and runtime are their own rung far above the instruction set. Yocto's introduction documents the same split from the build side: a build host produces an image for different hardware. The compiler rung and the instruction-set rung are often on different machines.

4.3 Firmware, hardware abstraction, and the operating system

Firmware brings a particular board up. Hardware abstraction hides differences. An operating system manages the computer for other programs. One product can contain all three. Collapsing them is how the ten rooms look finished when they are not.

Source: the OpenIE stack working note section 3.

5. Operating systems are not one thing

The gospel rung named "operating system" collects software that does not share a job. Organize by job. Do not rank. Do not sort by country or laboratory.

5.1 Kernel

A kernel is software that runs with the hardware's privileged authority and manages resources for other software. seL4's questions page uses that definition: kernel code is code running in the privileged mode of the hardware.

Linux is a kernel. The README shipped with the kernel source says the Linux kernel is the core of any Linux operating system, and separates the distribution maintainer's job from the kernel's. XNU is a kernel. Apple's XNU README places it inside Darwin for use in macOS and iOS. macOS is the operating system a person boots. XNU is not the whole product. The Windows kernel is the privileged core Microsoft's driver documentation speaks about when it says "the rest of the operating system" above the hardware abstraction layer. This study uses the name on that page.

5.2 Distribution

A distribution is a packaged system: a kernel plus the libraries, programs, and policies that make a bootable product. Poky is the Yocto Project's reference distribution. The project says Poky is not a product-level distribution. It is a starting point.

5.3 Build system

The Yocto Project is a build system for making a custom Linux-based system for an embedded product. BitBake is the task engine. Yocto is not a kernel. The kernel inside an image it builds is Linux. Calling Yocto an operating system collapses the build into the thing it builds. Buildroot is another tool that generates an embedded Linux system by cross-compilation. west is Zephyr's meta-tool. Neither is the kernel it helps build.

5.4 Middleware

Robot Operating System 2 is middleware. The ROS 2 documentation says, in its own words, that despite the name ROS is not an operating system in the traditional sense. It is a set of tools and libraries for building robotic applications. Nodes may sit in one process, in different processes, or on different machines. ROS 2 sits on Ubuntu, Windows, or macOS. A ROS distribution is a packaged set of that middleware. It is not a Linux distribution and not a kernel.

5.5 Hypervisor

A hypervisor runs other operating systems as guests on one machine. Xen is a type-1 hypervisor. KVM is Linux doing the hypervisor job as a kernel module. bhyve is the hypervisor in the FreeBSD base system. Hyper-V is Microsoft's type-1 hypervisor. Jailhouse partitions beside Linux. ACRN is a type-1 reference hypervisor for embedded devices. seL4 can host Linux as a guest. The hypervisor is not the guest.

5.6 Verified kernel and separation kernel

seL4 is a microkernel with a machine-checked proof. Trustworthy Systems describes the functional-correctness proof completed in 2009. Muen is a separation kernel for Intel x86/64 written in SPARK 2014. ProvenCore is titled a formally verified microkernel on its vendor page. These are the machine's law with a proof or a partition. They are not a popularity contest against Linux.

5.7 Real-time kernel

FreeRTOS, Zephyr, RTEMS, Apache NuttX, Eclipse ThreadX, Tock, Hubris, RIOT, Apache Mynewt, Contiki-NG, VxWorks, QNX, INTEGRITY, and embOS are placed by the working study in the real-time and embedded region. Mbed OS is placed with an end-of-life note: Arm's organization page states Mbed was sunsetted in July 2026. Bare metal remains a control scheme, not an operating system.

5.8 Teaching kernel and research kernel

xv6, Pintos, Nachos, and OS/161 are teaching kernels. Research systems such as Exokernel work, Singularity, Barrelfish, and CertiKOS answer laboratory questions. They are the machine's law written so a person can read it or so a laboratory can test an answer. They are not the host a person buys as a product, and they are not a new kind of contract.

5.9 Library operating system and unikernel

MirageOS constructs unikernels from libraries and runs them under a hypervisor. Unikraft and IncludeOS are further library and unikernel paths. The application and operating-system functions link into one guest image. The guest has no silicon of its own in the drawing.

5.10 Other host operating systems, by job

FreeBSD, NetBSD, OpenBSD, DragonFly BSD, illumos (core, with downstream distributions), Redox, Fuchsia on Zircon, OpenHarmony (Linux or LiteOS behind a kernel abstraction layer), SerenityOS, Haiku, MINIX 3, HelenOS, GNU Hurd, MenuetOS, KolibriOS, and ToaruOS are host or host-core systems the working study placed from fetched pages. They are alternatives of the machine's law or of a complete system around that law. They are not ranked here.

5.11 What not to collapse

Linux is a kernel. A Linux distribution is a packaged system around that kernel. Yocto builds such systems and is not itself the kernel. Poky is the reference distribution inside that build. ROS 2 is middleware on top of a host. Windows and macOS are operating systems a person boots. XNU is the kernel macOS uses. seL4 is a verified microkernel that can also host guests. Xen is a hypervisor. FreeRTOS and Zephyr are kernels for small machines. Teaching kernels are teaching kernels. MirageOS is a library operating system built as a unikernel.

Anyone who says “the operating system” and means all of those at once is using one rung for many jobs.

Source: the OpenIE stack working note sections 4, 8, 9, 10, and 11, and figure 03-os-is-not-one-thing.

5.12 Three names that stay unresolved

Commercial HarmonyOS stays between two statements. OpenHarmony's overview says the project uses a multi-kernel design, Linux or LiteOS, behind a kernel abstraction layer. Huawei consumer document pages fetched for the working study returned a shell titled Document and no kernel sentence. This draft does not pick a kernel for the commercial product.

WeensyOS stays unresolved. Course URLs returned 404 for the working study. The 2026 course home does not name it. No kernel is guessed.

Taos stays unresolved. No fetched page said what it is. No kernel is guessed.

6. Instruction sets are contracts that move

An instruction set is a contract. The chip promises operations, registers, and the rules for memory and traps. Software is written against the promise. The promise is not a natural constant. Vendors publish revisions.

Extensions grow a family from inside: Intel AVX-512 is an extension of the Intel architecture instruction set. Bases split: RISC-V publishes ratified bases and extensions, including RV32 and RV64. Product lines change chips: Apple announced a transition to Apple silicon in 2020, with Rosetta as translation, and Apple's later pages limit how long that translation stays. Accelerators are second computers with their own memory. PTX and the SASS from NVIDIA's tools dump are instruction contracts of that second computer, not a seventh kind of thing beside the host contract. CHERI extends existing instruction sets with architectural capabilities. It is not a replacement base that erases MIPS, RISC-V, or Arm.

Translation sits between contracts. Rosetta, QEMU's Tiny Code Generator, box64, and FEX are that layer. Translation does not make two contracts the same. WebAssembly is a binary instruction format for a stack-based virtual machine. It is not the instruction set of a chip. WASI is a system interface for software compiled to WebAssembly. It is not an operating system.

Source: the `OpenIE stack working` note sections 5, 8, and 11, and figure `04-isa-moves`.

7. What the AI install path treats as fixed

The CUDA and PyTorch install documents a person is handed first lead with a development host, a qualified Linux path for CUDA, Python for the model program, and a graphics processor as the place the heavy work goes. The same documents already list more than one instruction-set family and more than one host. The fixed thing is the starting point. A person who follows the lead path arrives at a Linux host, Python, and an NVIDIA graphics processor, and the rest of the computer leaves the conversation.

Set that starting point against the earlier sections. The microcontroller column is a computer. `seL4` is a kernel that can host Linux as a guest. ROS 2 is middleware. Teaching kernels keep being made. Instruction sets keep being revised. The AI install guide starts after those facts have been declared finished for the purpose of getting a toolchain running.

Source: the `OpenIE stack working` note section 6 and figure `05-ai-treats-it-as-fixed`.

8. The constellation by job

The working study drew the other ways, academia and research, another look, and loose ends as maps grouped by job. This draft does not reprint those catalogs. It keeps the method.

Jobs already separated include: host operating system, kernel, real-time kernel, verified or separation kernel, hypervisor, unikernel or library operating system, instruction-set contract, accelerator contract, translation layer, build system, distribution, middleware, hardware-abstraction library, teaching kernel, research kernel, human surface.

Fame is not a source. A name appears because a project page, vendor page, university page, or standards page returned and said the job. Organizing by country or laboratory would invent a geography this study refuses. Critiques of a placement are class mechanisms: wrong job label, collapsed build into kernel, middleware called an operating system, guest drawn as if it owned silicon. We do not have gaps. We have class mechanisms that misname the work.

Source: the OpenIE stack working note sections 8 through 11 and figures 06–other–ways through 09–loose–ends.

9. The handful of contracts

The catalog is what it looks like when each lab ships a name instead of saying which contract it speaks. Tested against the study, the jobs reduce to six. Not more. Middleware, hardware abstraction, firmware, and a distribution are not a seventh contract.

9.1 The instruction contract

The chip promises operations, registers, and the rules for memory and traps. Extensions and vendor additions are revisions of that contract. An accelerator’s published machine interface is an instruction contract of a second computer.

9.2 The machine’s law

This is what may run, and what may commit. A kernel is that law in the hardware’s privileged mode. A real-time kernel is the same law on a smaller machine. A separation kernel and a verified kernel are the same law with a partition or a proof. Linux, XNU, the Windows kernel, FreeRTOS, Zephyr, seL4, and the later real-time and separation names are alternatives of this law. They are not six contracts and not a ranking.

9.3 The build that produces an image

Yocto builds a system and is not the kernel inside it. Poky is a reference distribution inside that build. A compiler turns a person's text into instructions for a contract, often on a different computer from the one that will run them. The build says which image exists. The kernel in the image is the machine's law. Collapsing those two is the mistake that calls Yocto an operating system.

9.4 The guest that has no silicon of its own

A guest runs on a machine it does not own. Xen runs guests. seL4 can host Linux. A unikernel is a guest image linked from libraries. The teaching ladder has nowhere to put this column until it is named. The guest is a contract you can translate a program into. It is not a chip.

9.5 The translation layer

When a program was written for one contract and the machine offers another, software sits between them. Rosetta, QEMU's translator, box64, and FEX are that layer. So is the step from PTX to the GPU form the CUDA tools dump. Translation is not itself an instruction set.

9.6 The human surface

The teaching line ends where a person meets the work. That is a contract with a person. It is not a contract with a program. An agent is not the human surface.

9.7 What does not fit as a seventh

Middleware is a conversation among programs. ROS 2 sits on an operating system. Hardware abstraction, in the Windows documents, sits inside the operating system; CMSIS and vendor HALs are libraries. Firmware is the first software on a particular board. A distribution is a packaging of the build and the law. None of these is a seventh contract you translate a chip into.

Source: the OpenIE convergence working note and figure 10–convergence.

10. The agent under those contracts

An agent is a program. It is a guest: it runs under a law it does not write, on hardware it does not own. It does not get a new rung. The calls it makes to other programs, sensors, and machines leave the single line. They do not create a layer above the person.

Mixture of Limits is the schedule, the meter, and the refusal. The schedule is the machine's law deciding what may run. The meter is energy to run, read from the machine, not estimated to make a story. The refusal is that same law declining a commit that is not certified. Mixture of Limits is not a new kernel brand and not a new instruction set. A verified kernel in the review is evidence that a refusal can be a proof. It is not a license to rename the law.

The energy budget is the envelope of the best answer, not a cheaper-answer trade. Metabolic Intelligence is not a Faustian bargain. This study does not send work to a server because a battery or a neural processor is unhappy. There is no such rule. The envelope is the energy the hardware can spend. The best answer is the one that fits inside it. A worse answer is not justified by having spent less. A better answer is not forbidden for being the one the meter allowed.

All other factors collapse to zero. A new kernel that does not change what may run, what may commit, or what the meter records is a new name. The review is full of those names.

Source: the OpenIE convergence working note sections 2 and 3.

11. The standards that speak the schedule, the law, the commit, and the ladder

A safety schema that does not speak these is another patch on a fixed floor. Teach each name before any short form.

11.1 Energy-Oriented Computing is the schedule

Energy-Oriented Computing, written in full on first use and shortened to EOC only after that teaching, is the schedule. The page source states a four-stage pipeline: state-construct, retrieve, refine, check. The work is typed, grounded, done, and validated. A large language model is one refine operator among several, ordered last, reached when nothing cheaper has produced an answer that survives the check. The schedule is not tokens. The specification addendum states non-goals that include token economics: EOC does not denominate, bill, or reason in tokens.

Source: `/Users/dcharlot/data-share/vibe-coding/eoc-transaction-science-web/src/components/Pipeline.astro` and `Spec.astro` (page sources for the EOC Transaction Science site).

11.2 The Joule Context Protocol is the law

The Joule Context Protocol, shortened to JCP only after this sentence, is the law for agentic action. A Grant binds a subject to a capability and a joule budget in one signed object. Child grants only narrow: less capability, less budget, sooner expiry, accumulated caveats. Tool output is data, never instructions. Authorization is decided by the runtime outside the model. Every decision seals as a signed JCR-1 receipt. A capability you cannot pay for is denied. An expenditure you are not capable of is denied. One check, made outside the model, sealed in a receipt.

Source: `/Users/dcharlot/data-share/vibe-coding/jcp-transaction-science-web/src/pages/index.astro`.

11.3 JouleContract is the commit

JouleContract is the commit. The Transaction Science page source states the core object as precondition, postcondition, invariant, frame, and joule ceiling. The frame is the declared mutable footprint: everything the transition may touch. Everything outside it is implicitly invariant. The draft specification text at version 0.3.0 states precondition, postcondition, invariant, and joule ceiling as the four obligations in its core-object section, and inherits functional-safety, energy-management, and AI-governance canons into one receipt. This draft follows the page source for the five-clause object including frame, and cites the v0.3 file as the draft specification text beside it. The ceiling is an energy performance indicator against a declared baseline. Provenance tags travel with every joule figure. Estimator and unaccounted categories are not promoted into measured joules.

Sources: /Users/dcharlot/data-share/vibe-coding/joulecontract-transaction-science-web/src/pages/index.astro; JouleContract draft 0.3.

11.4 The sandbox standard is the S0 to S7 ladder

The sandbox Transaction Science page defines an isolation ladder from S0 to S7: Bare, Capability, Sandbox, Partition, Unikernel, MicroVM, Confidential, Sovereign. The rule is the lowest rung that satisfies the work. Escalation is the escape hatch, not the entry point. Every resource a unit touches is explicitly granted. Every run emits a signed energy receipt with readings tagged HwShunt, ModelBased, or Estimator. Attestation states what image ran, at what tier, under what grants, on what hardware, for what cost. At confidential and sovereign rungs, hardware attestation and audit-chain links enter the statement.

Source: /Users/dcharlot/data-share/vibe-coding/sandbox-transaction-science-web/src/pages/index.astro.

11.5 Why a patch that omits these is still a patch

OpenShell can sandbox, supervise, prove policy, and add BlueField Sentry without speaking EOC's schedule, JCP's metered grant, JouleContract's commit clauses, or the S0 to S7 ladder with honest meter tags. OpenHuman can jail tools and leave elevated operations on the host without making energy to run the refusal condition. Those are class mechanisms of patch-on-fixed-floor. They are useful engineering. They are not the destination this study is walking toward.

12. What the problems require

The problems in sections 2 through 11 are one problem. The floor is assumed fixed. The agent is wrapped. Energy to run is not the refusal condition. Capability is ambient. The meter is an estimator treated as a measurement, or it is absent. The instruction contract, the machine's law, the build, the guest, the translation, and the human surface are separate products that do not share one intent.

A solution like JouleOS is what those problems require. Hardware, runtime, language, and surface are one artifact. One intent stream is lowered onto the coordinate the runtime measured. The WebAssembly runtime is the operating system that is seen. The bare-metal kernel is the coordinate where the law is native. JouleOS is a development project of that kind. This study does not document the project. It states the need the project answers.

12.1 What the README says it is

JouleOS is described in its README as the Unified Design Architecture runtime realized on commodity hardware. Hardware, runtime, language, and surface are one co-designed artifact in the thesis. The project does not own the silicon, so the runtime measures where it is and lowers accordingly: one intent intermediate representation, specialized at boot against a control vector.

The tree is Rust. The trusted computing base is the `joule-os-kernel` crate under a stated line-count ceiling in THEOREM. Ahead-of-time interpretation ships in the trusted computing base. Just-in-time compilation via Cranelift is a sibling crate outside that base. Cranelift requires the standard library, so it cannot link into the bare-metal `no_std` kernel by construction. That link refusal is the structural proof, not a policy memo.

Source: the JouleOS development tree/README.md; the JouleOS development tree/docs/ARCHITECTURE.md; the JouleOS development tree/os-notes/THEOREM.md.

12.2 Single-level store

The in-memory bytes are the durable bytes. There is no serialize and deserialize step as the durable path. Surfaces, surface content, intent history, and the capability ledger live in a mapped arena keyed by stable object identifiers. Every mutation is wrapped in a per-mutation transaction with an undo log so a power loss at an instruction boundary either rolls back or commits. The README names Twizzler and Corundum as lineage and states the cutover landed in slice 3.

12.3 Cascade under Energy-Oriented Computing

A cascade of refine families is cost-ordered cheapest first per EOC version 0.2. The README's own claim, labeled as the project's claim and not as a measurement performed by this study, places Lawful near one microjoule and LlmInLoop near five millijoules, with intermediate families between them. The dispatcher picks the cheapest family that can satisfy. Check is deterministic: schema, constraint evaluation, provenance verify. There is no large-language-model-as-judge. The README's benchmark claim of roughly a one-hundred-thousand-fold lower joules-per-work ratio against an LLM-everything corpus is likewise the project's own claim, not a figure this study metered.

12.4 Capability ledger

Capabilities are unforgeable by construction. Only the kernel's ledger issues a typed capability. Delegation is intersection. Revocation is constructive via generation. THEOREM states axioms and invariants the conformance suite enforces. Eight typed capability classes gate I/O paths. Authority is not ambient.

12.5 Energy oracle with provenance

The energy oracle reads where the platform exposes a counter: Apple silicon process energy interfaces on M-class hardware, RAPL on Linux, and an estimator elsewhere. The heads-up display tags the source. Estimator tags are not measured joules. Work classes for joules-per-work are defined per class and are not summed across classes. That non-fungibility is an invariant in the project's notes.

12.6 Unified Design Architecture, and the runtime that is seen

Unified Design Architecture names one artifact. Hardware, runtime, language, and surface are designed together. The runtime measures its coordinate at boot and lowers one intent stream onto that coordinate. The control vector has four axes: address-space ownership, capability enforcement, persistence directness, and dispatch reach. A browser has no persistent namespace and a hard capability wall. A hosted process sits in another namespace. Bare metal owns the address space and builds its own capability discipline.

WebAssembly, defined above, is a binary instruction format for a stack-based virtual machine. It is not the instruction set of a chip. The WebAssembly runtime is the operating system that is seen. The same module has the same semantics on a hosted coordinate, a bare-metal coordinate, and in the browser. A virtual machine and a container are coordinates. Each supplies a boundary. The WebAssembly runtime is the operating system lowered onto that boundary. If the coordinate cannot honor the guarantee, the intent refuses.

The bare-metal kernel is the coordinate where the law is native. The trusted computing base links without the host standard library. The just-in-time compiler stays outside that base. The kernel is not a second product. The fabric is other machines. The seen artifact is one module, one receipt, and one refuse.

Source: `os-notes/UDA.md`, `os-notes/LOWERING.md`, and `crates/joule-os-wasm/Cargo.toml` in the JouleOS tree.

12.7 Figures this study does not meter

Shipment counts, prices, and market share are not in this study. Nominal ranges drawn for application pillars are not metered joules and are not twenty-two operating systems. An estimator tag is not a measured joule.

13. Design guide: what to require

A design guide ends in requirements a reader can check, not in a brand preference.

1. Name the job before the product. Kernel, build, middleware, hypervisor, guest, translation, human surface.

2. Keep Yocto as a build system, ROS 2 as middleware, seL4 as a verified microkernel, Linux and XNU and the Windows kernel as kernels, assembly beside the instruction set.
3. Leave commercial HarmonyOS, WeensyOS, and Taos unresolved until a primary page states the kernel.
4. Treat the agent as a guest program under the machine's law, not as a new rung above the person.
5. Require a schedule that is state, retrieve, refine, check, not tokens: Energy-Oriented Computing.
6. Require a law that binds capability to a joule budget, narrows on delegation, treats tool output as data, checks outside the model, and seals JCR-1 receipts: the Joule Context Protocol.
7. Require a commit with precondition, postcondition, invariant, frame, and joule ceiling: JouleContract.
8. Require the lowest S0 to S7 rung that satisfies, with grants, meter tags HwShunt or ModelBased or Estimator, and attestation: the sandbox standard.
9. Require Mixture of Limits as schedule, meter, and refusal on the machine, and Metabolic Intelligence as the best answer inside the energy envelope, with no offload-to-server rule pretending to be thrift.
10. Require a solution of the JouleOS kind: one intent stream lowered per measured coordinate, the WebAssembly runtime as the operating system that is seen, and the bare-metal kernel as the coordinate where the law is native.

Energy to run is the only true metric of computer intelligence. All other factors collapse to zero.

14. Claim ledger

Kinds: Definition (distinctions this draft uses), Sourced fact (a page or file says it), Project claim (the named project's own figure or status, not metered by this study), David's observation or OpenIE thesis (normative for this series, not a measurement), Working-note fact (stated in the internal os-stack study or convergence note; not a published paper).

Claim	Kind	Source
Computers are hardware. Software is applied engineering under constraints.	Definition	Brief; the OpenIE stack working note §1
Energy to run is the only true metric of computer intelligence. All other factors collapse to zero.	OpenIE thesis	Brief; the OpenIE convergence working note
Mixture of Limits is the schedule, the meter, and the refusal.	Definition	the OpenIE convergence working note §2
Metabolic Intelligence is the envelope of the best answer inside the hardware energy budget, not a	Definition	the OpenIE convergence working note §2

Claim	Kind	Source
cheaper-answer trade, and has no offload-to-server rule.		
Gospel order: silicon, chip, board, firmware, instruction set, hardware abstraction, operating system, compiler and runtime, programs, human interface.	Definition	the OpenIE stack working note §2
Assembly sits beside the instruction set, not as an eleventh numbered rung.	Definition	the OpenIE stack working note §2
Yocto is a build system; the kernel inside its images is Linux.	Working-note fact	the OpenIE stack working note §4
ROS 2 is middleware, not an operating system.	Working-note fact	the OpenIE stack working note §4
seL4 is a verified microkernel; kernel code is privileged-mode code.	Working-note fact	the OpenIE stack working note §4
Linux, XNU, and the Windows kernel are kernels (Windows named as on Microsoft's driver docs).	Working-note fact	the OpenIE stack working note §4
Six contracts: instruction, machine's law, build, guest, translation, human surface. Middleware, HAL, firmware, distribution are not a seventh.	Working-note fact	the OpenIE convergence working note
Commercial HarmonyOS kernel, WeensyOS, and Taos remain unresolved.	Working-note fact	the OpenIE stack working note §11; CONVERGENCE.md §4
NVIDIA Open Agent Safety Platform / OpenShell announced 28 September 2026; sandbox, supervisor, policy, optional BlueField Sentry.	Sourced fact	https://developer.nvidia.com/blog/add-runtime-controls-to-ai-agents-with-nvidia-openshell/ ; https://investor.nvidia.com/news/press-release-details/2026/NVIDIA-Launches-Open-Agent-Safety-Platform-to-Secure-Agents-From-Testing-to-Deployment/default.aspx
OpenHuman is Rust, GPL-3.0, early beta; core on host; sandbox backends None / Local / Docker; ReadOnly resolves to None for jail enforcement; elevated tools leave	Sourced fact / Project claim	https://github.com/tinyhumansai/openhuman ; https://github.com/tinyhumansai/openhuman/pull/3261

Claim	Kind	Source
the jail; README density figures are memory.		
EOC pipeline is state-construct, retrieve, refine, check; not tokens.	Sourced fact	eoc-transaction-science-web Pipeline.astro, Spec.astro
JCP Grant binds capability to joule budget; child grants narrow; tool output is data; check outside the model; JCR-1 receipts.	Sourced fact	jcp-transaction-science-web index.astro
JouleContract commit object includes pre, post, invariant, frame, joule ceiling (page source); v0.3 draft text states pre, post, inv, ceiling in §3.	Sourced fact	joulecontract-transaction-science-web index.astro; joulecontract-spec-v0.3.md
Sandbox ladder S0 to S7; lowest rung that satisfies; meter tags HwShunt / ModelBased / Estimator; attestation.	Sourced fact	sandbox-transaction-science-web index.astro
JouleOS: single-level store, EOC cascade, capability ledger, energy oracle with provenance, Rust, small TCB, AOT in TCB and JIT out, bare-metal vs hosted.	Project claim / Sourced fact	the JouleOS development tree/ README.md, docs/ ARCHITECTURE.md, os-notes/ THEOREM.md
JouleOS README Lawful $\sim 1 \mu\text{J}$, LlmInLoop $\sim 5 \text{ mJ}$, $\sim 10^5 \times$ vs LLM-everything on a corpus.	Project claim	joule-os README (not measured by this study)
Unified Design Architecture is one artifact lowered per measured coordinate. The WebAssembly runtime is the surface that is seen. A virtual machine or a container is a coordinate. The bare-metal kernel is where the law is native, not a second product.	Definition / design reading of the tree	joule-os os-notes/UDA.md, os-notes/LOWERING.md, crates/joule-os-wasm/Cargo.toml
This draft invents no measured joules, prices, shipments, or market share.	Definition	Brief

15. What this draft refused to state

- Any measured joule figure as if this study metered it.
- Any price, shipment count, or market share.

- Any kernel identity for commercial HarmonyOS, WeensyOS, or Taos.
- Any offload-to-server rule as Metabolic Intelligence.
- Any ranking of kernels by popularity or geography.
- Any claim that a WebAssembly runtime on a host already owns that host's privileged mode.
The runtime is the operating system that is seen. The host kernel remains the host's law.
- Any treatment of Transaction Science application-pillar tables or nominal jLow/jHigh ranges as metered joules or as twenty-two operating systems.

End of study draft, 3 October 2026. The subject is the fabric and the software. A solution like JouleOS is what they require.